

EE 210L — Digital Circuits and Logic Design Lab

Lab 5: Adders and Comparators

Name: _____ Date: _____

Partner(s): _____ Section: _____

Submit this report on Canvas — see Canvas for the deadline and grading rubric.

Part 1 — Half Adder

A	B	V at S (V)	S	V at C (V)	C	Expected S	Expected C
0	0						
0	1						
1	0						
1	1						

Part 2 — Full Adder

Draw the full adder circuit you built. Label every gate and pin.



Figure 1: Full adder built from gates.

A	B	C _{in}	V at S (V)	S	V at C _{out} (V)	C _{out}	Expected S	Expected C _{out}
0	0	0						
0	0	1						
0	1	0						
0	1	1						
1	0	0						
1	0	1						
1	1	0						
1	1	1						

Part 3 — Four-Bit Addition (74LS283)

A (dec)	A (bin)	B (dec)	B (bin)	$\Sigma 4 \ \Sigma 3 \ \Sigma 2 \ \Sigma 1$	C4	Sum as decimal	4-bit result correct?
9	1001	6	0110				
9	1001	11	1011				

How does C4 tell you that a 4-bit sum has overflowed?

Part 4 — Adder/Subtractor

Draw how the XOR gates and the control line connect to the 74LS283.

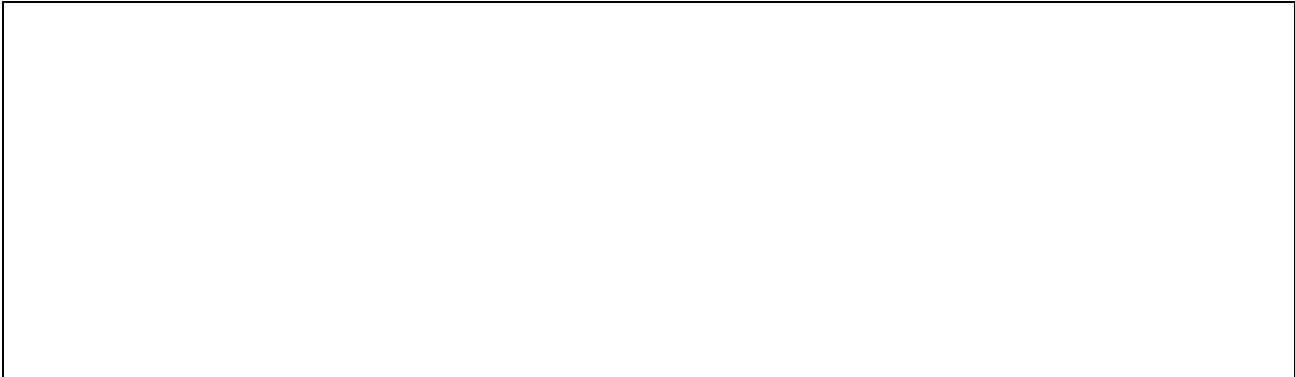


Figure 2: Adder/subtractor built around the 74LS283.

Control	Operation	A	B	$\Sigma 4 \ \Sigma 3 \ \Sigma 2 \ \Sigma 1$	C4	Result as decimal	Expected
0	A + B	7	3				
1	A – B	7	3				
1	A – B	12	5				
1	A – B						

What does C4 mean when the circuit is subtracting?
